

A feature fusion method based on radiomic features and revised deep features for improving tumor prediction in ultrasound images

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ABSTRACT

Background: Radiomic features and deep features are both vitally helpful for the accurate prediction of tumor information in breast ultrasound. However, whether integrating radiomic features and deep features can improve the prediction performance of tumor information is unclear.

Methods: A feature fusion method based on radiomic features and revised deep features was proposed to predict tumor information. Radiomic features were extracted from the tumor region on ultrasound images, and the optimal radiomic features were subsequently selected based on Gini score. Revised deep features, which were extracted using the revised CNN models integrating prior information, were combined with radiomic features to build a logistic regression classifier for tumor prediction. The performance was evaluated using area under the receiver operating characteristic (ROC) curve (AUC).

Results: The results showed that the proposed feature fusion method (AUC = 0.9845) obtained better prediction performance than that based on radiomic features (AUC = 0.9796) or deep features (AUC = 0.9342).

Conclusions: Our results demonstrate that the proposed feature fusion framework integrating the radiomic features and revised deep features is an efficient method to improve the prediction performance of tumor information.

1. Introduction

Breast cancer is a heterogeneous malignant tumor, and results in high death rates [1,2]. Early screening and diagnosis of tumor is helpful for the treatment of patients and reduction of mortality. Thus, it is vitally important to develop a prediction model for the diagnosis and subsequent treatment of breast tumor patients [3,4].

Medical imaging techniques, including ultrasound [5,6], magnetic resonance imaging (MRI) [7,8], and mammogram [9], provide a non-invasive method to assess tumor characteristic. Compared to other imaging technique, ultrasound has low cost and high repeatability [10–13], and has been widely used for early screening and diagnosis of breast tumor. Radiomics covered image to a large number of features, which reveal spatial heterogeneity of tumor [14–16]. Previous studies extracted radiomic features from breast tumor on ultrasound image to investigate the associations with pathological information of tumor, such as benign and malignant lesion [17–19]. Nemat extracted 855 radiomic features from the tumor region, and subsequently selected

optimal features to construct a logistic regression for predicting benign and malignant tumors [17]. The results of the above studies showed that radiomic features extracted from the tumor region are helpful for the accurate prediction of tumor information. However, the feature value rely on the tumor region manually annotated by radiologists. In addition, these features reflect superficial tumor characteristic, and high-level semantic information are lost.

In recent years, convolutional neural networks (CNNs) can automatically capture imaging features from images, and are widely used in classification of breast ultrasound image [6,19–22]. In a previous study [20], several CNN models, including VGG16, VGG19, InceptionV3, and ResNet50, were used for the diagnosis of breast cancer. The results of these studies showed that the CNN method presented superior performance for breast tumor classification. However, superficial information obtained from tumor region are lost. Whether integrating radiomic features and deep features obtained from breast ultrasound image can improve the performance for tumor prediction remains unclear [23–27].

To this end, a feature fusion method integrating radiomic features

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and revised deep features extracted by the revised CNN model fusing prior information was proposed to improve the prediction performance of tumor information. The main contributions of this paper are summarized as below.

- (1) Radiomic features were extracted from the breast tumor region, and Five optimal radiomic features were determined.
- (2) Several useful CNN models were used to predict tumor information, and the CNN with best prediction performance was selected as the final feature extractor. Furthermore, a revised CNN model fusing prior information was proposed to obtain revised deep features.
- (3) Optimal radiomic features and revised deep features were fused and used to construct classifier to improve the prediction performance of tumor information.

2. Materials and methods

The proposed feature fusion framework is shown in Fig. 1. The main steps are as follows. First, image preprocessing was performed to obtain the signal intensity in the tumor region. Then, radiomic features were extracted from the tumor region, and optimal radiomic features were obtained based on Gini score. Next, inspired by a recent study [23], a revised CNN model integrating prior information was used to obtain revised deep features. Finally, the optimal radiomic features and revised deep features were fused to construct a logistic regression (LR) classification for predicting benign and malignant tumors.

2.1. Ultrasound dataset

A publicly available breast ultrasound dataset (<https://scholar.cu.edu.eg/?q=afahmy/pages/dataset>) was used in this study [28,29]. The dataset consists of 780 images from 600 female patients with age between 25 and 75 years old. The average image size is 500×500 pixels. These images can be divided into three categories, namely normal images, benign lesion, and malignant lesion. The number of these three types is 133, 437, and 210, respectively. Due to no mass region, radiomic features of normal image cannot be obtained. Thus, the normal images were removed. Finally, a total of 647 images, including 437 benign lesions and 210 malignant lesions, were obtained in this study. These images were randomly divided into a training-validation

set (452 images, 70 %) and a testing set (195 images, 30 %) where the stratified sampling method was adopted to ensure the same proportion for each category between training set and testing set. In addition, another publicly available ultrasound dataset (<https://qamebi.com/breast-ultrasound-images-database/>) was used to validate the robustness of the proposed feature fusion model.

2.2. Image preprocessing

In our breast ultrasound images, tumor segmentation mask of each slice was also provided. Based on original breast images and tumor mask, the signal intensity of the tumor region was easily obtained, and used to extract corresponding radiomic features for tumor prediction. Fig. 2 shows the obtained tumor region-based signal intensity.

2.3. Feature extraction based on radiomics and deep CNN

2.3.1. Radiomic feature extraction and selection

A large number of radiomic features, reflecting superficial information of tumor region, were extracted using a publicly available Python package Pyradiomics. These features are divided into three categories, including shape, statistical, and texture features (<https://pyradiomics.readthedocs.io/en/latest/>) [14,30,31]. A total of 102 features were obtained where the features number of three types is 9, 18, and 75. Specially, texture features consist of gray level co-occurrence matrix (GLCM), gray level dependence matrix (GLDM), gray level run length matrix (GLRLM), gray level size zone matrix (GLSZM), and neighboring gray tone difference matrix (NGTDM) features.

The features were standardized using Z-score method. Subsequently, feature selection methods were used to determine the optimal radiomic features (ORFs). In our study, five-fold cross-validation method was used to determine the optimal ORFs based on Gini score.

2.3.2. Deep feature extraction

CNNs were used to extract deep features reflecting high-level semantic information of images. VGG16, VGG19, InceptionV3, ResNet50, ResNet101, ResNet152, DenseNet121, DenseNet169, EfficientNetB3, EfficientNetB4, and EfficientNetB5 models perform well in image classification tasks, and are thus selected to extract high-level semantic information of images in this study. Details for the above CNN architecture refer to previous studies [20,32–34].

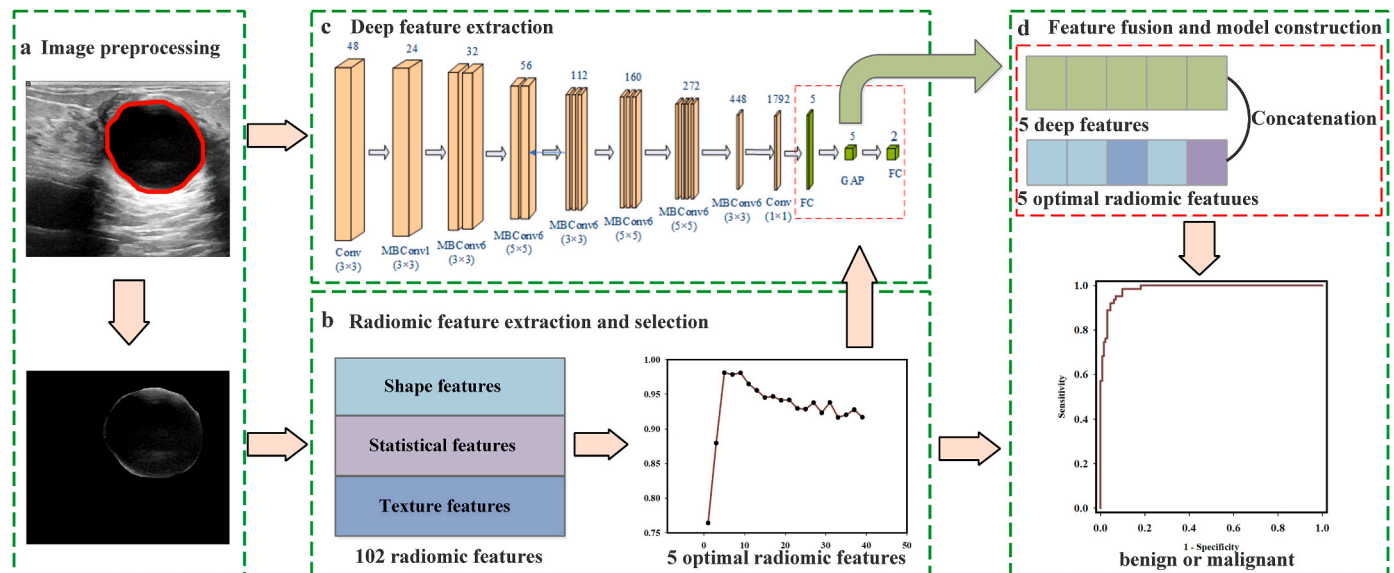


Fig. 1. Proposed feature fusion framework for breast cancer classification. (a) Tumor region. (b) Radiomic feature extraction and selection. (c) Revised deep features were extracted using a revised CNN model integrating prior information. (d) Feature fusion and classification model construction.

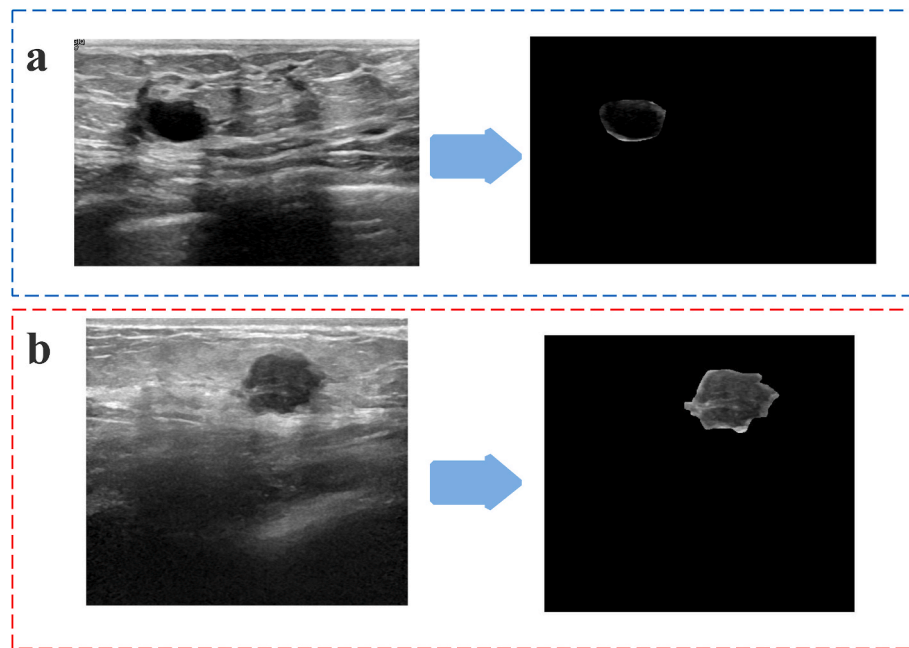


Fig. 2. An example of breast tumor region. (a) Benign tumor. (b) Malignant tumor.

The above CNN models were pretrained on the ImageNet dataset, and were subsequently fine-tuned using our breast ultrasound images based on transfer learning technique. Different from the radiomic features, the training-validation set were randomly divided into a training set and a validation set. The training set was used to train CNN models, and the validation set was used to determine the optimal hyper-parameters of CNN models. The used hyper-parameters included: batch size = 16, maximum epoch = 20, learning rate = 0.0001, and Adam solver. The CNN model with best prediction performance was selected to extract deep features for subsequent fusion.

The CNN model with best prediction performance was selected to extract deep features among the above CNN models. Furthermore, the optimal deep features were obtained based on the Gini score. In addition, inspired by a recent study [23], the revised CNN models integrating prior information were proposed to extract revised deep features where the neurons number of final fully connection layer is set to the number of the ORFs. To make it clear, the deep features extracted by the original CNN models, the optimal deep features, and the deep features extracted by the revised CNN models were respectively termed as the original deep features (ODFs), original optimal deep features (OODFs), and revised deep features (RDFs).

2.4. Feature fusion based on the ORFs and deep features

The ORFs and deep features were fused to improve the prediction performance of tumor. In this study, three feature fusion methods were included. First, the obtained ORFs and ODFs were concatenated, which is termed as the feature fusion method 1 (FFM1). Furthermore, to remove the redundancy of deep features, the ORFs and OODFs were concatenated, which is termed as the feature fusion method 2 (FFM2). To further improve the prediction performance of tumor information, the ORFs and RDFs were combined, which is termed as the feature fusion method 3 (FFM3).

2.5. Statistical analysis

Differences of the radiomic features and deep features were investigated between benign and malignant tumor group using Wilcoxon test. The prediction performance of individual feature was also evaluated.

Based on the fused features, a LR classifier was constructed to predict tumor information. The training set was used to train a LR classifier, and the testing set was used to evaluate the prediction performance of the classifier where the area under the receiver operating characteristic (ROC) curve (AUC) was used. For comparison, the ORFs, and deep features were also used to construct a LR classifier for tumor prediction, respectively. The prediction performance was compared between the feature fusion method and the ORFs- or deep features-based method using DeLong test. Additionally, the accuracy, sensitivity, and specificity were also reported. In this study, $p < 0.05$ was considered statistically significant differences.

3. Results

3.1. Performance based on ORFs for tumor prediction

3.1.1. Gine score-based feature selection

Based on the Gini score, 5 ORFs were determined to construct a LR classifier for tumor prediction. Table 1 lists the Gini scores of the ORFs where the shape feature of sphericity obtained the highest Gine score of 0.1203 among the ORFs.

3.1.2. Difference analysis of the ORFs

Fig. 3 shows the box plots of 2 ORFs. The shape feature of sphericity and perimeter showed significant differences between benign and malignant tumors. Additionally, the shape feature of sphericity obtained

Table 1
Univariate analysis based on 5 ORFs.

	Feature name	Gini score	<i>p</i> value	AUC	Accuracy
Top 1	Sphericity (shape)	0.1203	3.9840e-35	0.8401	0.7949
	Perimeter (shape)	0.0424	1.5171e-27	0.8010	0.7282
Top 2	Run entropy (texture)	0.0362	3.2107e-25	0.8051	0.7436
Top 3	Elongation (shape)	0.0310	1.8326e-06	0.6494	0.6667
Top 4	Energy (Statistical)	0.0264	1.5315e-19	0.7292	0.6667
Top 5					

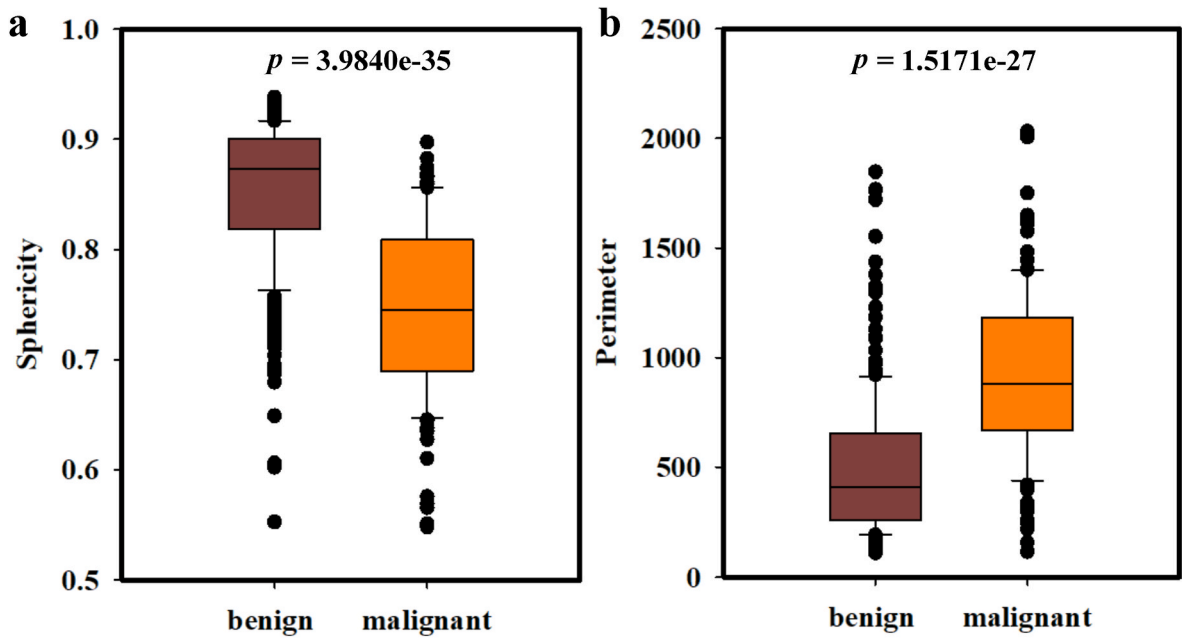


Fig. 3. Box plots of 2 ORFs extracted from the tumor region. (a) The shape feature of sphericity. (b) The shape feature of perimeter.

the highest AUC of 0.8401 and accuracy of 0.7949 among the ORFs (see Table 1).

3.1.3. Multivariate analysis for tumor prediction

5 ORFs were used to construct a LR classifier for predicting tumor benign and malignant tumors. The LR classifier achieves a high AUC of 0.9796 and accuracy of 0.9077. The results indicate that the selected ORFs are vitally helpful for the tumor prediction.

3.2. Performance based on deep features for tumor prediction

3.2.1. Prediction performance of classical CNN models

Table 2 lists the prediction performance of the classical CNN models. The EfficientNetB4 model obtained the highest AUC of 0.9449 and accuracy of 0.9026 among the CNN models. Additionally, the sensitivity and specificity are 0.8889 and 0.9091, respectively.

Table 3 lists the prediction performance of the revised CNN models. The revised EfficientNetB4 model obtained the highest AUC of 0.9522 and accuracy of 0.8872 among the CNN models. Specially, the revised EfficientNetB4 model obtained a higher AUC value compared to the original EfficientNetB4 model (see Tables 2 and 3). Additionally, the sensitivity and specificity are 0.9206 and 0.8712, respectively.

3.2.2. Analysis of deep features obtained from the original EfficientNetB4 model

Based on the results of the CNN models, the EfficientNetB4 model

Table 2

Prediction performance of classical CNN models.

CNNs	AUC	Accuracy	Sensitivity	Specificity
EfficientNetB4	0.9449	0.9026	0.8889	0.9091
EfficientNetB5	0.9238	0.8564	0.7460	0.9091
EfficientNetB3	0.9253	0.8769	0.7778	0.9242
DenseNet169	0.8773	0.8256	0.7302	0.8712
DenseNet121	0.8911	0.8256	0.6825	0.8939
ResNet101	0.9309	0.8564	0.8095	0.8788
ResNet152	0.8968	0.8154	0.4762	0.9773
ResNet50	0.9245	0.8256	0.5238	0.9697
InceptionV3	0.9071	0.7538	0.254	0.9924
VGGNet19	0.794	0.7333	0.6667	0.7652
VGGNet16	0.8479	0.7744	0.6825	0.8182

Table 3

Prediction performance of the revised CNN models.

CNNs	AUC	Accuracy	Sensitivity	Specificity
EfficientNetB4	0.9522	0.8872	0.9206	0.8712
EfficientNetB5	0.9259	0.8615	0.8413	0.8712
EfficientNetB3	0.9255	0.8872	0.8254	0.9167
DenseNet169	0.9065	0.8667	0.7460	0.9242
DenseNet121	0.9112	0.8462	0.7460	0.8939
ResNet101	0.9354	0.8769	0.8571	0.8864
ResNet152	0.9118	0.8462	0.6984	0.9167
ResNet50	0.9366	0.8769	0.7460	0.9393
InceptionV3	0.9325	0.8410	0.8095	0.8560
VGGNet19	0.8505	0.8154	0.7302	0.8561
VGGNet16	0.8909	0.8051	0.8254	0.7955

was selected and used as an extractor for obtaining deep features. Finally, a total of 1792 deep features were extracted. Based on the Gini score, 35 OODFs were obtained and used for subsequent analysis. Table 4 lists the Gini scores of 5 OODFs where Deepfeature1062 obtained highest Gine score of 0.0309.

Fig. 4 shows the box plot of 2 OODFs. The Deepfeature1062 and Deepfeature215 both showed significant differences between benign and malignant tumors. Among 5 OODFs, the Deepfeature1062 obtained the highest AUC of 0.8401 (see Table 4).

Table 4

Univariate analysis based on top 5 OODFs.

	Feature name	Gini score	p value	AUC	Accuracy
Top 1	Deepfeature1062	0.0309	2.1848e-46	0.8401	0.8103
	Deepfeature215	0.0294	2.0452e-43	0.7992	0.7385
Top 2	Deepfeature1381	0.0227	4.4667e-50	0.7713	0.7538
Top 3	Deepfeature1632	0.0209	7.6409e-44	0.8316	0.8205
Top 4	Deepfeature303	0.0178	1.1000e-36	0.7420	0.7538
Top 5					

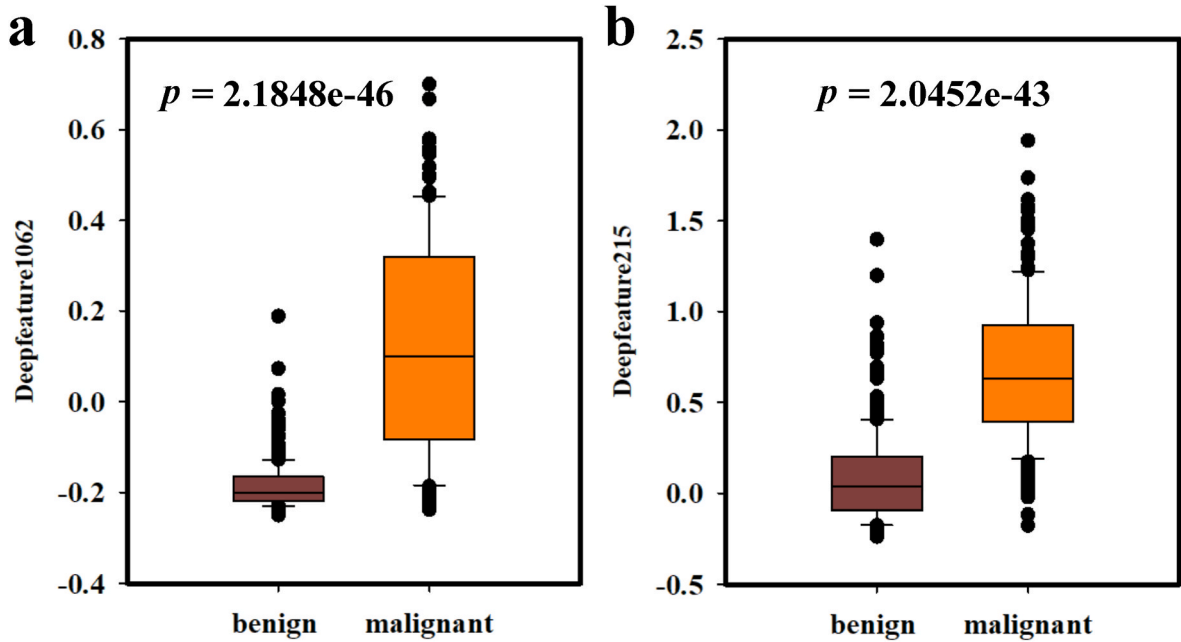


Fig. 4. Box plots of 2 OODFs. (a) Deepfeature1062. (b) Deepfeature215.

3.2.3. Analysis of revised deep features obtained from the revised EfficientNetB4 model

To be consistent with the ORFs [23], the number of final neurons of the fully connection layer was set to 5. Table 5 lists the p values of 3 RDFs obtained by the revised EfficientNetB4 model. Fig. 5 shows the box plots of Deepfeature1. The Deepfeature1 showed significant differences between benign and malignant tumors.

3.3. Performance based on the fusion of ORFs and deep features

5 ORFs were combined with 1792 ODFs and 35 OODFs obtained from the original EfficientNetB4 model, respectively. Finally, 5 ORFs and 3 RDFs obtained from the revised EfficientNetB4 model were combined to improve the prediction performance of tumor.

Table 6 lists the prediction performance of the fused features integrating the ORFs and revised deep features. Fig. 6 shows the ROC curves of the fused features, ORFs and deep features. The FFM3 obtained higher AUC of 0.9845 and accuracy of 0.9385 than the framework based on the ORFs or deep features. Similar findings were obtained on our ultrasound dataset with simulated segmentation error and an additional ultrasound dataset. The detailed results are listed in Supplementary Tables S1 and S2. Additionally, the sensitivity and specificity were 0.9524 and 0.9318, respectively. Table 7 lists the DeLong p values between two different methods. Specially, the FFM3 significantly improved the prediction performance compared to the FFM1 or the deep feature-based method.

Additionally, our results also showed that the ORFs obtained a higher AUC of 0.9796 and accuracy of 0.9077 than the deep features. Specially, significant improvement was also found between ORFs and the deep features (see Table 7).

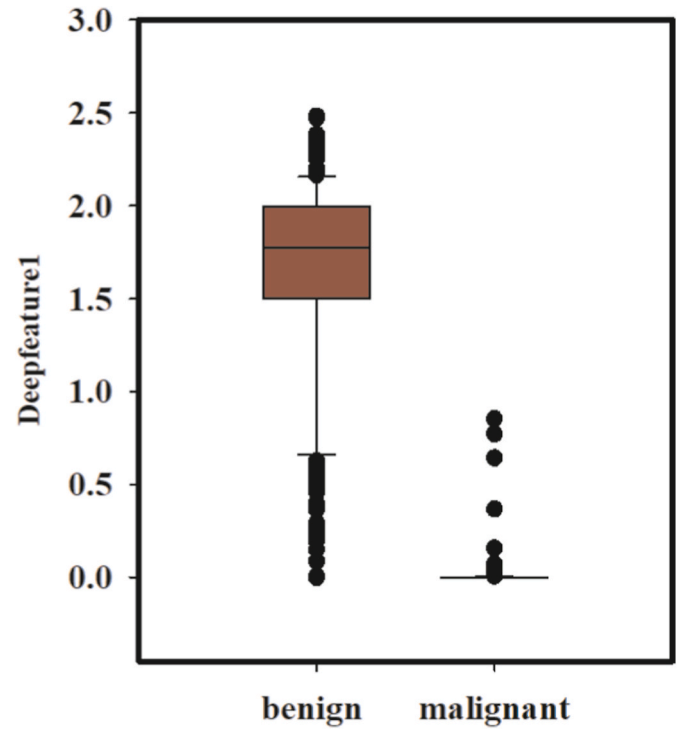


Fig. 5. Box plots of Deepfeature1.

Table 5
Univariate analysis based on 3 RDFs.

Feature name	p value	AUC	Accuracy
Deepfeature1	2.2501e-63	0.9342	0.8205
Deepfeature2	1.8529e-63	0.9345	0.8205
Deepfeature3	2.0297e-63	0.9342	0.8205

Table 6
Prediction performance based on the feature fusion framework.

Method	AUC	Accuracy	Sensitivity	Specificity
FFM3	0.9845	0.9385	0.9524	0.9318
FFM2	0.9672	0.9282	0.8889	0.9470
FFM1	0.9212	0.8769	0.8571	0.8864
ORFs	0.9796	0.9077	0.8730	0.9242
RDFs	0.9342	0.8256	0.9365	0.7727
OODFs	0.9190	0.8564	0.8095	0.8788
ODFs	0.9171	0.8718	0.8571	0.8788

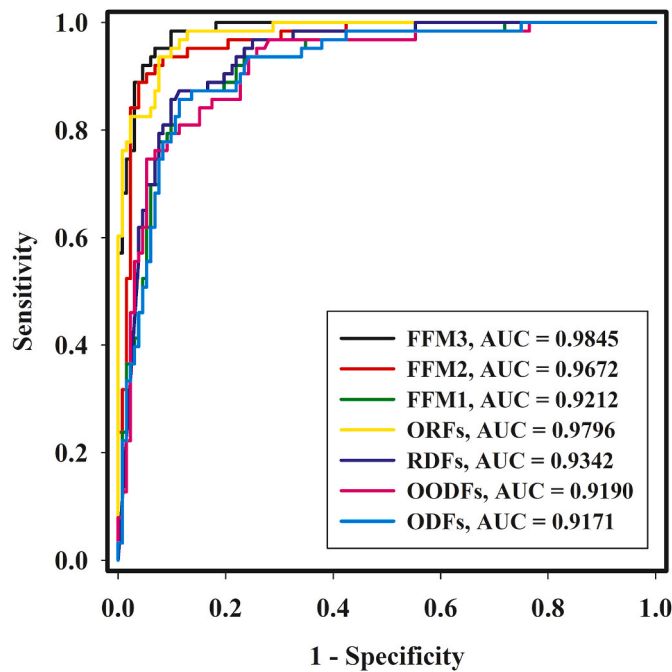


Fig. 6. ROC curves of the fused features, ORFs and deep features.

Table 7

Significance analysis (p value) between two different methods.

	FFM1	RDFs	OODFs	ODFs
FFM3	0.0010	0.0010	0.0013	0.0007
	0.0015	/	0.0003	0.0010
FFM2	/	/	/	0.0014
FFM1	0.0040	0.0133	0.0046	0.0029
ORFs				

4. Discussion

This study proposed a feature fusion framework to improve prediction performance of tumor. Our results showed that the feature fusion framework obtained better performance compared to the ORF- or deep feature-based method. To a certain extent, the finding supports that the radiomic features and deep features are complementary and thus improve the prediction performance of tumor [10,23,24].

Previous studies demonstrated that radiomics is an efficient tool to explore the associations between imaging feature and clinical information [17,18]. In this study, 102 radiomic features, including shape, statistical, and texture features, were used to investigate the associations with tumor information, and 5 ORFs were finally selected. These features are useful and can be directly used to build prediction models in subsequent ultrasound studies. Additionally, our results also showed that the ORFs consist of 3 shape features, 1 texture features, and 1 statistical feature. The results indicate that shape features make a greater contribution to the tumor prediction compared to texture or statistical features.

In addition to radiomics, CNN is also an efficient method and can be used to automatically extract useful deep features for accurate prediction of tumor information [21,35]. In our work, the prediction performance of several classical CNN models was investigated for tumor information. Our results showed that the EfficientNetB4 model obtained better prediction performance than the other CNN models. The finding indicates that the EfficientNetB4 model is vitally helpful for tumor classification in ultrasound images. In addition, our results also showed

that the prediction performance of the deep features was significantly lower than that of the ORFs. The results indicate that the radiomic features are more efficient than the deep feature obtained from the CNN models for tumor prediction. A possible reason is that the limited number of samples is not enough to train CNN model well, and useful deep features closely associated with tumor information are lost.

Different from previous studies [32], our studies assumed that redundant features were included among 1792 deep features, and important features should be obtained for improving prediction performance of tumor. Our results showed that the OODF-based method obtained higher AUC than the ODF-based method. In addition, our results showed that the FFM2 obtained the better prediction performance than the FFM1. These findings support the assumption reported in related studies [36,37].

Previous studies predict tumor information by integrating the radiomic features and deep features, and the results showed the fused features are useful for tumor prediction [23,26]. To this end, this work proposed a feature fusion method integrating the ORFs and revised deep features to improve the prediction performance of tumor. Our results showed that the FFM3 obtained better prediction performance compared to the ORFs or deep features. Specially, significant differences were found between FFM3 and FFM1 or deep features. The findings indicate that the feature fusion method can improve the prediction performance of tumor, which further supports the findings of previous studies [10].

There are several limitations in this study. First, the samples are small for CNN training. In future studies, more breast ultrasound dataset would be collected to train CNN model well and improve the prediction performance of these models. Second, this study only focused on the analysis of ultrasound image. In the future, the other imaging techniques, including MRI and mammogram, would be combined with ultrasound images to improve the prediction performance of tumor information [38,39]. Third, as a preliminary study, classical CNN models were used to extract deep feature and predict tumor information. In the future, more efficient CNN models would be adopted to extract more useful deep features. Finally, our study is based on the 2D image. In future study, a 3D image-based feature fusion framework would be proposed to extend the current work [40].

5. Conclusions

This study proposed a feature fusion model integrating the radiomic features and revised deep features to improve the prediction performance of breast ultrasound tumor. Our results demonstrated that the proposed feature fusion method is an effective method to accurately predict tumor information. These findings are helpful for the accurate diagnosis of breast tumor patients and determination of appropriate treatment plans.

CRediT authorship contribution statement

Xianyang Wang: Validation. **Linlin Lv:** Methodology. **Qingfeng Tang:** Investigation. **Guangjun Wang:** Supervision. **Enci Shang:** Methodology. **Hang Zheng:** Methodology. **Liangliang Zhang:** Methodology.

Ethics statement

This retrospective study was approved by the Institutional Review Board.

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Declaration of competing interest

The authors have no relevant conflicts of interest to disclose.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compbiomed.2024.109605>.

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